

Energy efficiency in cold supply chains of the food Sector: An exploration of conditions and perceptions

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ABSTRACT

Temperature-controlled cold supply chains (CSCs) are essential to ensure the availability of high-quality food products all year round. Cooling and freezing account for approximately 30 % of the electricity consumption in the food sector alone. Improving energy efficiency could therefore make a substantial contribution to achieving climate neutrality in this sector. Many analyses of energy efficiency seek to understand the adoption of energy efficiency measures (EEMs) in individual companies. Yet they hardly address the adoption of EEMs specifically in CSCs, in particular in the food sector. The novelty of this paper lies in analysing the conditions for and perceptions of energy efficiency along food CSCs. This also covers the implications of applying the theory on barriers to EEM in such chains. For this purpose, the paper combines the results of an interview study with 61 participants and a survey with 122 respondents from private sector organizations active in food CSCs in the European Union. It focuses on factors affecting the adoption of EEMs in CSCs, in particular a) the structure of food CSCs, b) non-energy benefits and c) behavioural and organizational aspects. The findings suggest that a) energy efficiency is more prevalent in individual companies than along entire CSCs, that b) non-energy related benefits seem relevant but are less perceived along CSCs and that c) the adoption of EEMs in CSCs is complex due to the structural, behavioural and organizational factors only present in multi-company settings such as CSCs. Linking these findings to the theory of barrier underlines that it is advisable to explicitly consider the inherent characteristics of CSCs when investigating barriers to energy efficiency and deriving implications for policy making thereof.

1. Introduction

With regard to limiting global warming to 1.5 °C (United Nations, 2015), a shift towards climate neutrality is needed in Europe before 2050 (European Commission, 2018). The widespread deployment of energy-efficient and climate-friendly technologies is therefore required across all sectors. An important demand-side sector is the food, beverage and tobacco industry, which accounted for a 12 % share of both final energy consumption and electricity demand in EU27's industry in 2019 (EUROSTAT, 2021). With about 30 % of the sectoral electricity demand, cooling and freezing (Monforti-Ferrario et al., 2015) are important end-uses. Increasing the adoption of energy efficiency measures (EEMs) in this area could thus make a significant contribution towards climate neutrality.

Temperature-controlled cold supply chains (CSCs) make it possible to transport perishable food products across large distances and offer them independently of the season. A supply chain, in general, can be defined as a set of three or more organizations linked by upstream or

downstream flows of goods from the producer through intermediate links to an end customer (Brzozowska et al., 2016; Monczka, 2009). A CSC, as a specific type of supply chain, describes the interdependent processes and equipment used to ensure the temperature preservation of perishable goods from source to final recipient in a safe manner and maintaining good quality (Zhang, 2007). A food cold chain infrastructure, as a particular type of CSC, includes pre-cooling facilities, cold warehouses, refrigerated carriers, packaging and traceability measurement tools (Shashi et al., 2018). This is a complex process due to the strict hygiene and refrigeration requirements involved that strongly influence energy consumption and the product quality offered to final consumers.

Various studies have investigated the cooperation of companies in the area of sustainable or green supply chain management (Carter and Rogers, 2008; Elliott, 2013; Seuring and Müller, 2008; Srivastava, 2007), particularly in the food sector (Govindan, 2018; Li et al., 2014; Murphy and Adair, 2013). It is generally perceived that green supply chain management promotes efficiency and synergy among business

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partners and contributes to all three dimensions of sustainable development, i.e. economic, environmental and social aspects (triple bottom line approach, see [Elkington, 1994](#)). Greening the supply chain to enhance environmental performance, minimize waste and achieve cost savings requires supply chain partners to collaborate upstream with raw material suppliers as well as downstream with final consumers ([Li et al., 2014](#)). However, various inter-organizational barriers seem to be prevalent in multi-tier supply chains, as pointed out in a review by [Govindan et al. \(2021\)](#) in the specific context of social sustainability.

The literature on energy efficiency, in particular, discusses the adoption of measures to improve energy performance usually independently of the specific considerations of CSCs. A key observation is that there appears to be general underinvestment in EEMs, also described as the “energy-efficiency gap” ([Jaffe and Stavins, 1994](#)). It is rooted in orthodox economic theory that describes how, under certain assumptions, competitive markets lead to a maximization of social welfare. The deviation from this optimum in terms of the “gap” is explained by the concept of barriers to energy efficiency, i.e. mechanisms that inhibit “a decision or behaviour that appears to be both energy efficient and economically efficient” ([Sorrell et al., 2004](#)). Seeking to explain this observation with concepts from other areas of economic theory, [Sorrell et al. \(2004\)](#) developed a well-established taxonomy of barriers to energy efficiency which draws on elements from orthodox and agency theory, behavioural economics and transactions cost economics. This resulting taxonomy distinguishes risks, imperfect information, hidden costs, access to capital, split incentives and bounded rationality as major categories of barriers to energy efficiency. Based on this and related classifications, barriers to energy efficiency have been investigated in many works (e.g. [Smith et al., 2021](#); [Sorrell et al., 2010](#); [Thollander et al., 2020](#)). Other studies, which are rooted more firmly in the social sciences than economic theory, stress the role of behaviour and the interaction of individuals within organizations (e.g. [Palm, 2009](#)). In recent years, more attention has also been paid to the role of non-energy benefits from energy efficiency (e.g. [Cooremans, 2011](#); [International Energy Agency, 2014](#)). Non-energy benefits (NEBs) can be described as improvements due to energy-efficient technologies that yield “additional enhancements to the production processes” ([Worrell et al., 2003](#)). Such enhancements include improvements in areas such as waste generation, emissions, operation and maintenance, production and working environment. It has been pointed out that NEBs can have a significant impact on the value of EEMs, even exceeding energy savings alone (e.g. [Thema et al., 2019](#); [Worrell et al., 2003](#)).

These studies of energy efficiency and the related theory on barriers concentrate on individual companies, not on EEMs within CSCs. Such measures can be perceived as action that requires the active engagement of at least two companies operating in the CSC to improve energy efficiency (e.g. energy-optimized delivery and storage schedules; harmonization of storage and transport temperatures; electricity-supply for trucks with electric cooling units; contracts for sharing of costs and benefits). Only a few investigations deal with the particular role and potentials of such energy efficiency across several companies of a supply chain or its integration into supply chain management ([Marchi and Zanon, 2017](#)). In a review on supply chain management with a particular focus on sustainability and energy efficiency, [Centobelli et al. \(2018\)](#) identify several research gaps and questions related to these two topic areas, but without further distinction for them. When it comes specifically to the improvement of energy efficiency in cold chains of the food sector, a classification of different possible cold chain stages has been suggested ([James and James, 2010](#)). There, EEMs are investigated with an emphasis on technological issues, while the collaboration between the different stages is not considered. Another study proposes an analytical model that jointly looks at economic and energy aspects of the food supply chain ([Zanon and Zavanella, 2012](#)). As a key factor of chain optimization, it considers specific requirements in terms of temperature and storage time that preserve the product quality over time, support decision-makers and improve the sustainability of the supply chain.

In sum, there currently seem to be no works specifically dedicated to the conditions for and perception of energy efficiency within CSCs in the food sector and the implications for the theory on barriers to energy efficiency. This paper aims to contribute to closing this gap. In particular, it seeks to explore them by providing insights into CSC structure, by analysing the relevance of behavioural and organizational aspects (BOAs) related to energy efficiency and by investigating the relevance of NEBs in CSCs. Based on an empirical basis, it intends to contribute to the empirical and theoretical literature by analysing the general conditions for energy efficiency improvements in this sector along the following questions:

- What role does energy efficiency play in CSCs of the food sector compared to its role in individual companies?
- What challenges do the structural features of CSCs pose for energy efficiency?
- What is the relevance of NEBs and how are they perceived in CSCs compared to individual companies?
- What are the particular challenges stemming from BOAs with regard to energy efficiency improvements along CSCs?
- What are the implications of these findings for the theory on barriers to energy efficiency?

The starting point for the analysis is a review of the literature (Section 2), followed by a description of the methodology (Section 3). The results are then presented (Section 4) and a discussion of these results based on the research questions (Section 5) before conclusions are drawn (Section 6).

2. Literature review

2.1. Structural features of (food cold) supply chains

Historically, direct-to-consumer deliveries and sales at farmers' markets represented the basic models of food supply. Due to the growth of the food processing sector and industrialization processes, intermediaries were added to supply chains and transportation functions expanded ([Chiffoleau et al., 2016](#)). Highly globalized supply chains, so-called mainstream agro-industrial models, with logistically complex mass distribution systems are an important part of today's market, while more traditional and innovative forms of food sales, such as local or short supply chains, are re-emerging, particularly in food markets ([Majewski et al., 2020](#)).

With regard to the **structure of food supply chains**, various scientific studies agree on the complex and dynamic relationships of the heterogeneous actors involved ([Mikkola, 2008](#)). A supply network typically includes vertical, dyadic relationships between a company and a single or multiple customers and suppliers, and horizontal relationships between suppliers ([Centobelli et al., 2018](#)). There are distribution chains of varying organizational structures and lengths ([Majewski et al., 2020](#)). Long industrial chains, in particular, are often complex and rationally organized networks ([Marsden et al., 2000](#)). The different types of food supply chains, from pick-your-own to sales to hypermarket chains ([Malak-Rawlikowska et al., 2019](#)), can be distinguished using different parameters, such as the product's destination, the number and roles of intermediaries involved, and the type of product ([Cadilhon et al., 2003](#)).

Since the 1990s, alternative food networks in industrialized nations have been studied that question the mainstream agro-industrial concept ([Chiffoleau et al., 2016](#)). Evidence suggests that local supply chains perform differently than mainstream supply chains. Differentiating between three **supply chain types** (mainstream, direct market and intermediated), a series of coordinated case studies in the U.S. compared the structure, size and performance of local food supply chains with those of mainstream supply chains ([King et al., 2010](#)). The basic structure of **mainstream supply chains** - characterized by distribution

centres that receive products from many suppliers and efficiently distribute a wide range of products to supermarkets - is effective and highly adaptable. In contrast to mainstream supply chains, **local food supply chains** convey detailed information about where and by whom products were produced and have relatively few segments that are often linked by trading partner relationships characterized by high degrees of trust and information-sharing. Seasonality creates a link between the two types: some mainstream supply chains obtain products from local farmers during certain times of the year and from national or international farmers in the off-season. **Intermediated supply chains** are highly diverse. They have the potential to aggregate local products so that they can be processed, distributed and/or marketed in sufficiently large quantities to achieve economies of scale. From an economic perspective, producers in local food supply chains tend to receive higher per-unit revenues and retain a larger share of the retail price. However, producers in most local supply chains assume greater responsibility for supply chain functions (e.g. processing, distribution, and marketing), which can be costly. Durable relationships between supply chain partners over time are important in all three types of supply chains (King et al., 2010).

From an ecological perspective, **food distribution systems** related to the transportation of goods are of special interest. For example, short food supply chains¹ (SFSCs) are commonly associated with reduced energy consumption, among other reasons, due to their organizational distribution and the shortened physical transportation distance between primary producers and final consumers. There seems to be a consensus on the economic and social benefits (close cooperation, geographic and social relationships between producers, processors and consumers) that can be attributed to SFSCs (Majewski et al., 2020). However, the results of an empirical study of 428 short and long European food supply chains suggest that shortening food supply chains in terms of organizational proximity does not automatically entail a more environmentally sustainable alternative to long supply chains nor mitigate some of the negative environmental externalities. The analysis, based on a life cycle assessment (LCA) approach, showed that the environmental impacts of the food distribution process are not only determined by the geographical distance between producer and consumer, but depend on numerous factors, including the supply chain infrastructure (Majewski et al., 2020).

Beyond the environmental impacts of food transportation, there do not seem to be any specific scientific studies of the **impact of food (cold) supply chain features on energy efficiency improvements**. While cold chain associations address sustainability issues in published grey literature (e.g. in Germany: Diekmann et al., 2015; Eberle et al., 2016), they do not publish data on their members or the structure of CSCs. A literature review of supply chain management for improved energy efficiency identifies the evaluation of supply chain features for energy performance improvements as one stream of research (Marchi and Zanoni, 2017).

To conclude, the highly heterogeneous character of the different food supply chains is evident in the literature. Understanding these structures is important since these may affect the adoption of EEMs in the CSCs of the food sector, as they determine information flows, transaction costs, risk perceptions and other factors. It therefore seems relevant to understand structural features such as the dimensions of supply chains, the origins of the companies involved, their interactions, familiarity with

each other and the long-term stability of CSCs.

2.2. Behavioural and organizational aspects

With regard to the adoption of EEMs, research on BOAs is strongly related to the research on **barriers to energy efficiency** (DeCanio, 1998; Trianni et al., 2013). Andrews and Johnson (2016) concluded after a review of the literature that most of the barriers are neither technical nor economic but behavioural.

Since supply chain cooperation is seen as a major opportunity to overcome barriers to implementing EEMs (Marchi and Zanoni, 2017), some studies specifically address the multi-actor relations present in supply chains. A conceptual framework suggested by Banks and Redgrove (2012) covers several levels of action that can shape a company's behaviour with respect to energy, including the relationships it has with other companies in its supply chain. The review of Centobelli et al. (2018) on the crucial role of energy efficiency in supply chain management indicates that the adoption of energy efficiency and environmental sustainability initiatives is mainly driven by relational (e.g. customer expectations) and regulation factors (e.g. government support). The highlighted barriers that hinder energy efficiency initiatives in supply chains include blockage barriers (e.g. investment costs), contingency barriers (e.g. company size) and market barriers (e.g. market volatility). Besides technical factors, such as tracking systems, **behaviour- and organization-related factors** that positively/negatively influence energy efficiency in supply chains are being elaborated for future research. These include human and cultural factors (e.g. competence and knowledge), company factors (e.g. organizational size) and social skills (e.g. cooperation), among others (Centobelli et al., 2018). In particular, the communication item emerges as one central supply chain construct to obtain more sustainable development (Khalid et al., 2020). In this context, ten key generic factors have been identified in a systematic review of collaboration behavioural factors for sustainable agri-food supply chains as well as other types of supply chain to broaden the basis of investigation (Dania et al., 2018): joint efforts, sharing activities, collaboration value, adaptation, trust, commitment, power, continuous improvement, coordination and stability. The authors highlight that joint efforts can help to overcome existing limitations in the system that lead stakeholders to focus on their own operations rather than forming a cooperative system due to geographic difficulties, poor support systems or a lack of resources, for example. However, the interplay among these factors as well as their impacts on achieving sustainability measures are strongly affected by the characteristics of products, regulations, environmental conditions and the socio-economic context involved (Dania et al., 2018).

With regard to **coordination schemes in supply chains**, these can either be centralized, i.e. one person is in charge of the decision-making process managing the whole supply chain, or decentralized, i.e. multiple decision-makers with conflicting objectives are involved which may lead to an inefficient system (Jaber and Goyal, 2008). This is why supply chain management is singled out as one of the main ways to overcome existing barriers (Marchi and Zanoni, 2017). However, in a case study of three Finnish vegetable supply chains, it was observed that economic activities seem to be strengthened by duplex or even multiplex relations, when compared to being managed by a single coordinative form (Mikkola, 2008). Furthermore, one great difficulty within supply chain coordinative relations and structures seems to come from their dynamic, invisible and possibly confidential nature (Mikkola, 2008).

The missing collaboration among supply chain members is also stressed in the context of **decision-making in cold chain logistics** (Chaudhuri et al., 2018). Here, the lack of monitoring and control along the cold chain and the missing exchange of temperature data are pointed out. To ensure the adoption of data analytics across the chain, it is important to identify the players in the chain who drive and coordinate the effort (Chaudhuri et al., 2018). Complementing this, a review of **food cold chain management** lists factors that can negatively affect

¹ The European Rural Development Regulation (1305/2013) defines a short supply chain as one that "has a limited number of economic operators, committed to co-operation, local economic development, and close geographical and social relations between producers, processors and consumers" (Official Journal of the European Union, 2013). The specific feature of SFSCs is the existence of a limited number (usually a maximum of one) of intermediaries between the farmer (primary producer) and the final consumer (Majewski et al., 2020).

efficient and effective cold chain management (Shashi et al., 2018). Besides organizational factors related to the technical cold chain infrastructure, such as improper traceability, behavioural factors are included, such as the irregular flow of information between a large number of intermediaries and a lack of expertise.

To conclude, behavioural and organizational factors seem to play a crucial role in driving energy efficiency. According to Marchi and Zanoni (2017), understanding the causes and patterns behind them is key to successfully improving energy efficiency. Interactions between organizations in the supply chain have proven highly influential. Given the complexity of CSCs in the food sector, it follows that BOAs could be of particular relevance for the adoption of EEMs.

2.3. Non-energy benefits of energy efficiency

When solutions to reduce energy consumption in a sector are considered, a common principle of appraisal - regardless of the specific method used - is to compare the costs of implementation with the resulting savings in energy costs. However, several studies indicate that the strategic value of energy efficiency in providing a competitive advantage can have a substantial impact on whether energy efficiency investments are made (e.g. Cooremans, 2015).

From a strategic perspective, in addition to the evident energy savings, **non-energy related effects**, such as enhanced competitiveness, reduced maintenance requirements or improved working conditions, are important factors affecting the assessment of EEMs. These so-called non-energy benefits (Mills and Rosenfeld, 1996) or “multiple benefits” (Ryan and Nina, 2012) have been investigated in several studies (e.g. International Energy Agency, 2014). Earlier studies examined the

productivity benefits of industrial EEMs. They identified substantial NEBs and categorized them using a taxonomy comprising six areas (Worrell et al., 2003): reduced waste, lower emissions, improved maintenance and operating costs, increased production and product quality, an improved working environment, and others such as improved public image or worker morale (not in order of importance). Later work aimed at a broader consideration of NEBs by introducing a framework that balances both non-energy benefits and non-energy losses (Cagno et al., 2019). It has also been pointed out that such benefits are more persuasive when linked to the values and priorities of decision-makers (Fawcett and Killip, 2019).

Studies using cost-benefit analysis suggest that the value of these benefits can even be higher than direct energy cost savings, with monetised ‘non-energy’ effects up to several times the magnitude of the energy cost savings (Worrell et al., 2003). By exceeding the monetary value of energy savings, NEBs affect the return on investment and help reduce costs, generate value, and mitigate risk (Cooremans, 2011).

To conclude, the literature on NEBs shows that a strategic assessment of co-benefits has the potential to positively influence decision-makers to invest in EEMs. However, there are only a few empirical studies available on the practical implementation in companies and none specifically concerning CSCs in the food sector. Furthermore, to the best of our knowledge, there are no studies examining the topic in detail beyond the level of individual companies. As this could be relevant for promoting supply chain energy performance, it seems particularly interesting to compare perceptions of NEBs in individual companies with the supply chain as a whole.

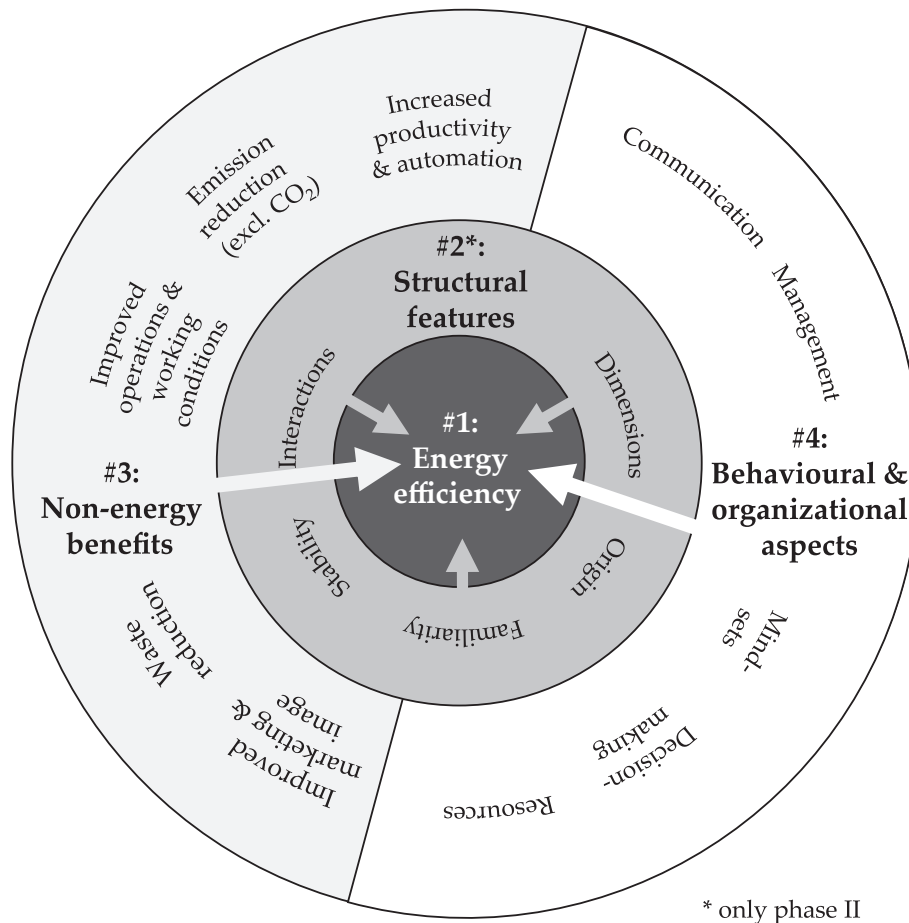


Fig. 1. Conceptual framework: The focus is on energy efficiency improvement, which is affected by the structure of CSCs. The perceptions of non-energy benefits as well as behavioural & organizational aspects that influence the adoption of energy efficiency measures also depend on these structures.

3. Methodology

The literature review highlighted numerous factors affecting the adoption of EEMs in CSCs of the food sector. The research design (Fig. 1) chosen to explore this issue assumes that the role of energy efficiency (#1) is affected by the structural features of CSCs (#2), the perceptions of NEBs (#3) and the role of BOAs (#4). Each of these areas includes various factors that may influence the adoption of EEMs.

Two phases of investigation were carried out to explore these factors. The first phase consisted of explorative interviews for orientation (phase I), while the second phase was a targeted survey (phase II) for additional substantiation. The explorative interviews focused on the role of energy efficiency (#1), NEBs (#3) and BOAs (#4); structural features (#2) were added only to the survey to avoid overburdening the interviews (Fig. 1). The target group of both phases consisted of representatives from companies or associations of the food sector operating in different stages of the CSC (production and processing, storage and logistics, wholesale and retail) and familiar with energy and sustainability topics (e.g. energy managers) and/or the CSC of the food sector.

3.1. Phase I: Interviews

Phase I consisted of semi-structured, in-depth stakeholder interviews to gain insights into the perception of energy efficiency along CSCs (see also Neusel et al., 2020). The interviews were explorative and combined both open and closed questions using a guideline to facilitate and harmonize them. Regarding NEBs, the interviews sought to gain insights into the perspective of both single companies and CSCs. Due to their particular focus on inter-company interaction within CSCs, questions were only asked about BOAs concerning the CSC. When analysing the results qualitative content analysis was applied: Answers to open questions were clustered by categorizing them according to a successively refined classification scheme. This recursive design allowed for open coding, i.e. new codes arising from new answers could be added (Saldaña, 2011). The number of thematic incidences was used as an indicator to determine thematic relevance, helping to synthesize the results of the large number of interviews.

In December 2019 and January 2020, a total of 61 interviews were conducted via telephone or face-to-face in 11 different countries. The interviews lasted between 15 and 85 min, with an average of 45 min. The major countries were Germany (16 interviews), Italy (15 interviews) and Spain (9 interviews). The majority of interviewees worked in companies (59 participants) and others (2) were employed in associations related to the food sector; 36 companies are active in production and processing, 11 in storage & logistics, 10 in wholesale & retail and the remaining 4 belong to refrigeration system and insulation suppliers and associations. A wide range of different sectors was covered, with the largest share from the meat sector (various: 18; meat: 16; dairy products: 7; fish: 5; fruit & vegetables: 3; beverage: 3; HORECA: 1; others: 8). Concerning company size, an almost even distribution was achieved between small, medium and large companies (small and micro: $n = 17$, medium: $n = 21$, large: $n = 21$).

3.2. Phase II: Survey

Phase II was a broader online survey conducted in eight languages. The target group were also food sector companies operating in different stages of the CSC across the EU. The aim of the anonymous survey was to substantiate and focus the preliminary results obtained from the interviews. In addition to the topics in the interviews, the survey also covered the structural features of cold chains. The survey used closed questions (partly multiple answers). Some questions could be skipped if the respondent lacked first-hand knowledge/experience of the area and some allowed multiple answers. In addition, some questions related first to the individual company as a reference and then to the CSC.

More than 1000 potential participants from food sector were invited,

either directly or via relevant associations. The survey was open for two months from April to June 2020. 150 evaluable results were obtained despite the first wave of the pandemic strongly affecting the sector. A review of answering behaviour indicated that some respondents limited their answers to the structure of the CSC. Assuming that these respondents were mainly knowledgeable about the CSC and not familiar with energy-related matters, their answers were included in case of a meaningful minimum of answers and duration in the survey.

A total of 122 evaluable surveys from private sector companies was used for this study. In terms of origin, the majority of respondents worked for organizations operating from Spain (30), Italy (28), Austria (19) and France (18). 72 respondents were active in production and processing, 25 in storage and logistics, and 18 in wholesale and retail (others: 7). A range of subsectors was covered, with a larger share of respondents from the meat sector (various: 37; meat: 34; fruit & vegetables: 14; dairy products: 10; HORECA: 7; fish: 4; beverage: 3; others: 13). There were similar shares of small, medium and large companies (small and micro: 37; medium: 38; large: 47).

Generally, it should be noted that representativeness cannot be ensured for CSCs in the food sector, neither for the interviews nor the survey results due to a general lack of data on representative CSCs in the food sector.

4. Results

The figures in this part result from a descriptive analysis of the survey results and indicate the number of valid answers for each item. As the upstream interviews were mainly explorative, statements and examples from them (see also Neusel et al., 2020) are used to underline the results. Deviations in the interviews and survey results are covered in the discussion.

4.1. Energy efficiency in the cold supply chain

With regard to the role of energy efficiency in **individual companies** as a reference, 44 % of the survey respondents stated that it is relevant in most or even all decisions, while another 50 % say that it is considered in at least some decisions (Fig. 2). Examples for EEMs include replacing old refrigeration equipment, improving insulation through cold curtains, using excess heat, introducing closed refrigeration units in retail stores, or modern refrigerated trucks. When asked about the relevance of energy efficiency from the perspective of the entire CSC instead, the answers were similar (Fig. 2), with most of the participants indicating that energy efficiency is considered in most or even all decisions (46 %) or at least sometimes (43 %) in the CSC. To put this into context, about 28 % also indicated that their company has a formal energy management system under ISO 50001 ($n = 33$), while about 63 % organize energy-related matters without a formalized energy management system ($n = 74$). Only a minority does not follow up on energy-related issues ($n = 10$) (no answer: $n = 4$).

According to the interviews, energy efficiency is considered relevant mainly due to the potential cost savings, which are necessary to ensure the competitiveness of products on the market by decreasing production costs. Despite this, a majority of those interviewed mention the typical obstacles to the introduction of EEMs (e.g. Sorrell et al., 2004) covered in literature. These include the high initial investments and long payback times of EEMs. A lack of information and know-how is also underlined. This also includes missing competences to optimize especially older refrigeration systems, too few qualified personnel to handle new machines and monitoring systems and a lack of information from the final customer. The problem is compounded by various issues including lack of awareness of energy efficiency amongst workers, who cause short-term interruptions of the cold chain by leaving doors of cooling equipment open, inefficient stock replenishment with partially filled storages, as well as customers with low willingness to pay more for the purchased product. Other mentioned obstacles include difficulties

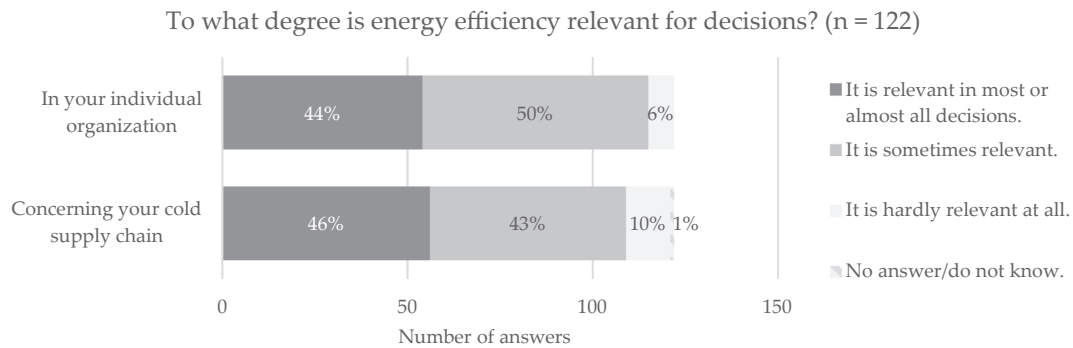


Fig. 2. Relevance of energy efficiency in decision-making processes of individual companies vs the cold supply chain.

with moving from existing to new systems (for instance the conversion to natural refrigerants according to the F-gases regulation or implementing measures during running operations) and the lack of the latest technology, adequate monitoring and maintenance systems. One obstacle that particularly addressed the cold chain logistic relations is the lack of communication and coordination. The different temperature ranges of food producers for their products (e.g. minced meat at 2 °C, sausage at 7 °C and cheese at 10 °C) mean that the cold chain is designed for the lowest temperature, i.e. 2 °C during transport and storage if required. Another obstacle that was often mentioned is the sub-optimal utilization and organization of logistics.

In terms of groups driving energy efficiency along the CSC, storage and logistics (31 %) as well as production and processing (28 %) are perceived as drivers, wholesale and retail seem hardly relevant (2 %) as a group (Fig. 3). On the other hand, there were also large shares of responses indicating that all the stages are equally relevant (21 %) and that no one pushes the topic for the whole CSC (18 %). It should be noted here that there were more survey respondents from production and processing (72) and storage and logistics (25) than from wholesale and retail (7) (see Section 3.2).

To sum up, respondents stated that energy efficiency is regularly considered in decision-making and there was only a small difference in the survey between decisions taken in individual companies and entire CSCs. Furthermore, storage & logistics as well as production & processing are indicated as important groups driving energy efficiency in the CSC.

4.2. Structural features of cold supply chains

A CSC spans several stages (“links”) such as farming, processing, distribution, retailing, local delivery, etc. In terms of supply chain dimensions, several companies can be active within each link e.g. due to multiple sourcing (Fig. 4). Nearly-two-thirds of the sampled CSCs span 4

to 9 links (34 % have 4–6 links and 32 % have 7–9 links; choice of categories as in Fig. 4). Concerning the number of companies in the chain, there is a large share with 1 to 10 companies (66 %); but large chains with up to 100 companies are also present (16 % with 41–100 companies). Regardless of the chain’s dimension, according to the survey results, at least half of the companies are often small or medium-sized enterprises (SMEs) with up to 250 employees.

In about half of the cases (47 %), the companies are typically located in the same country; 13 % even come from the same local area. The remainder operate in a larger geographical area, i.e. on the same continent (29 %) or globally (11 %) (Fig. 5a “origin”). Half of the respondents indicated that the companies in a CSC know their direct suppliers and customers (50 %). Another third (38 %) expected companies to know the next direct links to their customers and only 12 % seem to know all the companies involved from farm-to-fork (Fig. 5b “familiarity”). With regard to the stability of the chain, half of the respondents (51 %) stated that it has remained unaltered over several years, while roughly the other half (45 %) said there were a few newcomers and/or drop-outs each year (Fig. 5c “stability”). The frequency of other than purely routine interactions between companies in a CSC shows a heterogeneous situation: There are similar shares of cases with no contact at all outside routine interactions (18 %), with contact on a semi-annual (18 %) basis, a quarterly basis (16 %), a monthly basis (23 %) and cases with exchanges on a weekly or shorter basis (25 %) (Fig. 5d “interactions”).

To sum up, the analysis of structural features underlines the heterogeneous character of CSCs: While CSC members tend to be geographically close to each other and more likely to have long-term relationships, the number of companies per chain varies greatly, as does the frequency of interactions and familiarity with each other.

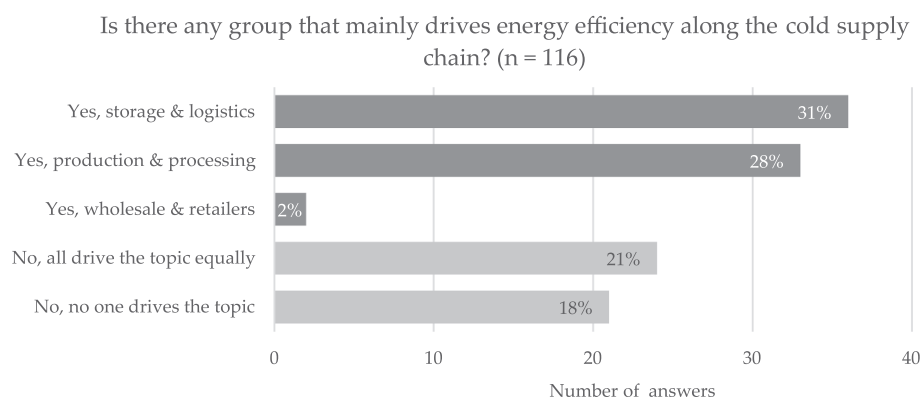


Fig. 3. Relevance of different stages of the cold supply chain for driving energy efficiency forward along the chain.

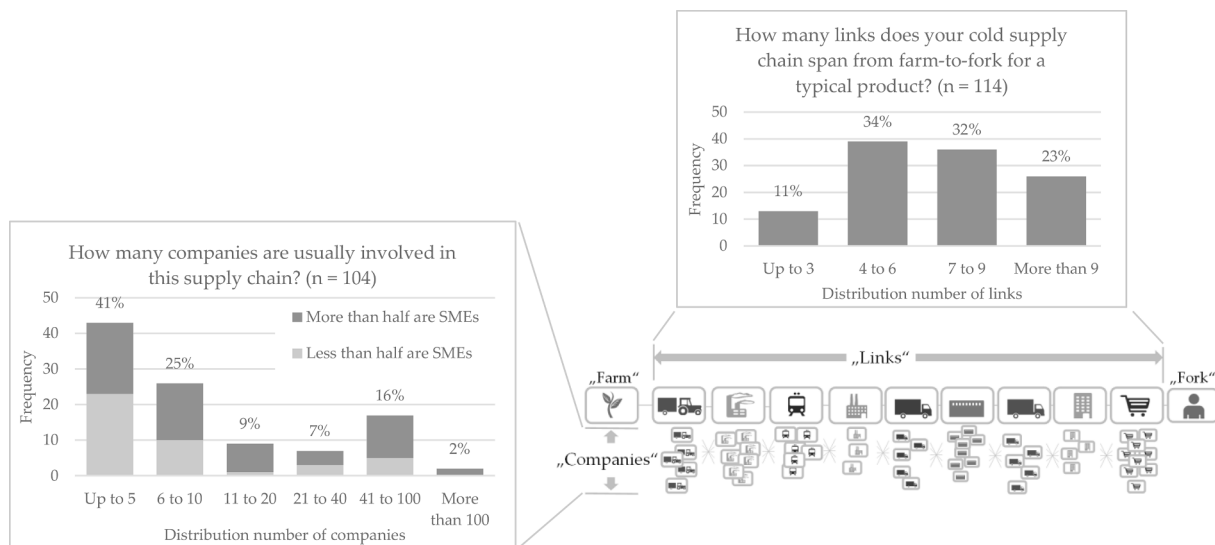


Fig. 4. Illustration of cold supply chain dimensions: number of links and companies in the analysed supply chains as well as the share of SMEs. A rather heterogeneous composition emerges.

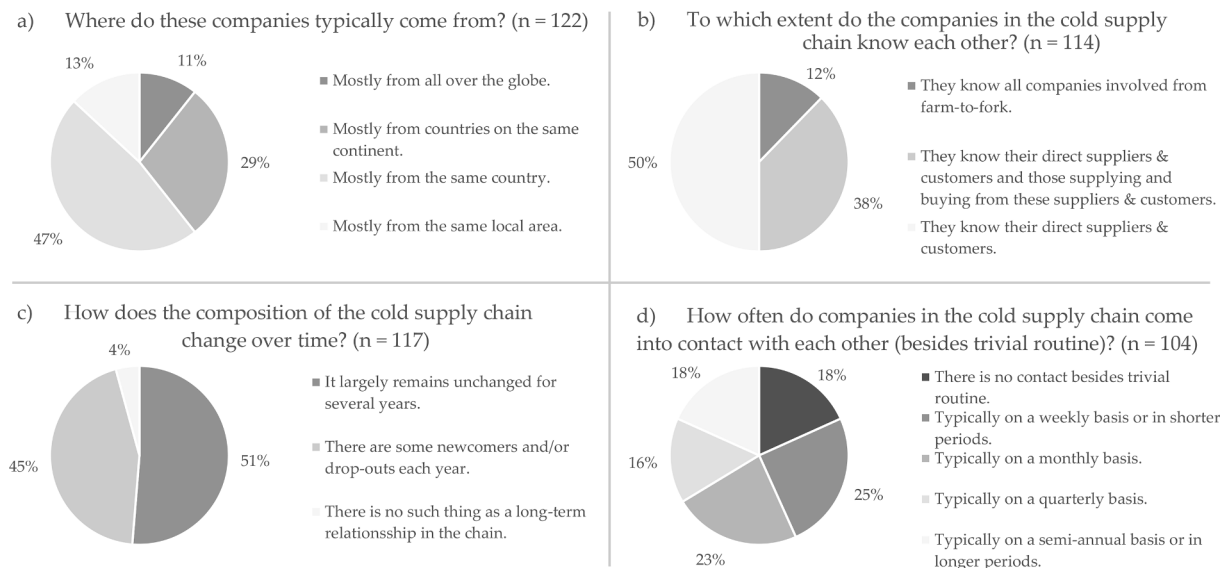


Fig. 5. Structural features of cold supply chains: a) origin; b) familiarity; c) stability; d) interactions.

4.3. Non-energy benefits in the cold supply chain

Concerning NEBs, respondents were invited to reflect on the positive or negative effects of recently implemented EEMs. The majority (69 %) of **individual companies** associated positive effects with implemented EEMs besides reducing energy demand and the related energy costs and carbon dioxide emissions (Fig. 6). 18 % stated they had observed neither positive nor negative effects, and 8 % had not implemented any EEMs recently. When asking the same question about the whole **supply chain**, the number of responses was lower, but the general perception was similar to the situation in individual companies. The majority seemed to perceive positive effects (32 %). Negative effects during or after the implementation of EEMs do not seem to be an issue for individual companies nor for CSCs as a whole. However, for the CSC, the percentage of those who observed neither positive nor negative effects (28 %) or had not recently implemented EEMs that affected other companies in the chain (16 %) was proportionally higher compared to individual companies.

To understand the relevance of benefits from EEMs in more detail,

respondents were asked to choose from a list of 6 categories of NEBs derived from Worrell et al. (2003) (multiple answers possible) – for their individual company and their supply chain (Fig. 7). From an individual company perspective, improved operations and maintenance was at the top of the list, closely followed by emission reduction (excluding CO₂) and improved working environment. From a supply chain perspective, emission reduction was mentioned most frequently, closely followed by better marketing and image, and waste reduction. It should be noted that the number of responses for positive effects on the supply chain was considerably lower. Furthermore, the sixth category of other potential NEBs was not particularly relevant for individual companies or CSCs.

To sum up, positive side-effects of EEMs outweigh any negative side-effects by far and seem to be relevant from the perspective of the companies and entire CSCs. However, the response rate for this topic is clearly lower for the CSC than for the individual company. There is also some evidence of a shift in emphasis: indirect positive factors, such as less waste and an improved image, were mentioned more frequently for the supply chain than for the individual company.

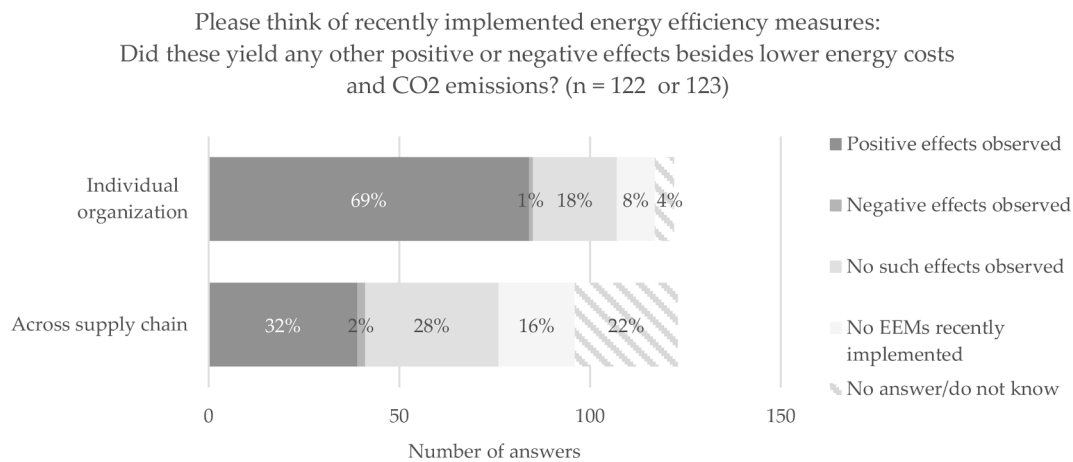


Fig. 6. Relevance of positive and negative side-effects of energy efficiency measures for individual companies vs the cold supply chain.

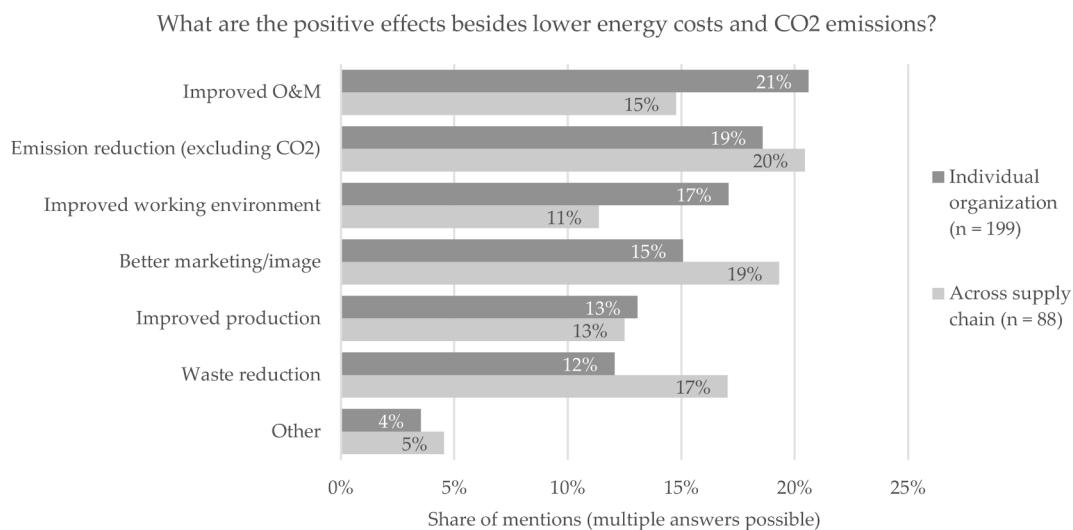


Fig. 7. Perceived non-energy benefits of energy efficiency measures for individual companies vs the cold supply chain.

4.4. Behavioural and organizational aspects along the cold supply chain

Respondents were asked about the role of BOAs in five categories: communication along the chain, decision-making processes, mind-sets of companies, the CSC management, and resources. Questions about BOAs were limited to respondents with first-hand experience of the exchange on energy-related topics along the CSC to avoid reporting hearsay: 45 % of respondents stated they had witnessed an exchange on energy-related topics along the CSC (Fig. 8, left graph).

There are relevant challenges to energy efficiency from all five categories. The specific challenges per category (Fig. 8, right table) underscore the findings of the preceding interviews: The companies find it challenging that there is often no or no regular **communication** between the different members of the CSC and no person in charge of overseeing the whole chain and coordinating the exchange. Communication only takes place within the scope of regulations and contracts in many cases, for example, between the quality assurance departments of producers and their logistic partners. If there are common **decision-making processes** related to energy efficiency among the cold chain members, frequently mentioned difficulties concern measuring the effectiveness and economic benefits of EEMs due to the lack of commonly available energy performance indicators. Barriers such as the diverging priorities of actors with different (technical) backgrounds, missing cooperation in terms of common financial resources and varying

regulation at country-level were also mentioned. With regard to **mind-set/behaviour**, the greatest challenge according to the participants is that the different actors do not feel responsible due to a lack of awareness and attention to energy-related topics ("never change a running system"). Different priorities as well as different perceptions of risk are reinforced by the cultural variety of the (international) actors. Organizational challenges in the chain's **management** occur due to a lack of cooperation and coordination, especially in the logistics stage. These include excessive cooling of some products, sub-optimal routing and waiting times, partially loaded trucks and non-optimal inventory levels in warehouses. In addition, many participants pointed to transparency issues in the CSC causing traceability challenges, such as sub-optimal temperature monitoring or storage planning. Dense dynamic regulation and administrative burdens were perceived as adding to the problem ("when companies are focusing more on passing the audits than on actually becoming more energy efficient"). **Resources** represent a major challenge for individual companies as well as for the CSC, where respondents saw the lack of a sustainable financing solution for the whole network. Financing EEMs was not always seen as easy due to short payback requirements. Immediate profits often come first so that decision-makers cannot easily grasp the economic benefits of investing in EEMs in the long run. This is accompanied by a lack of know-how and missing experts or engineers in the field of refrigeration, which calls for additional resources for education and training. Data availability was

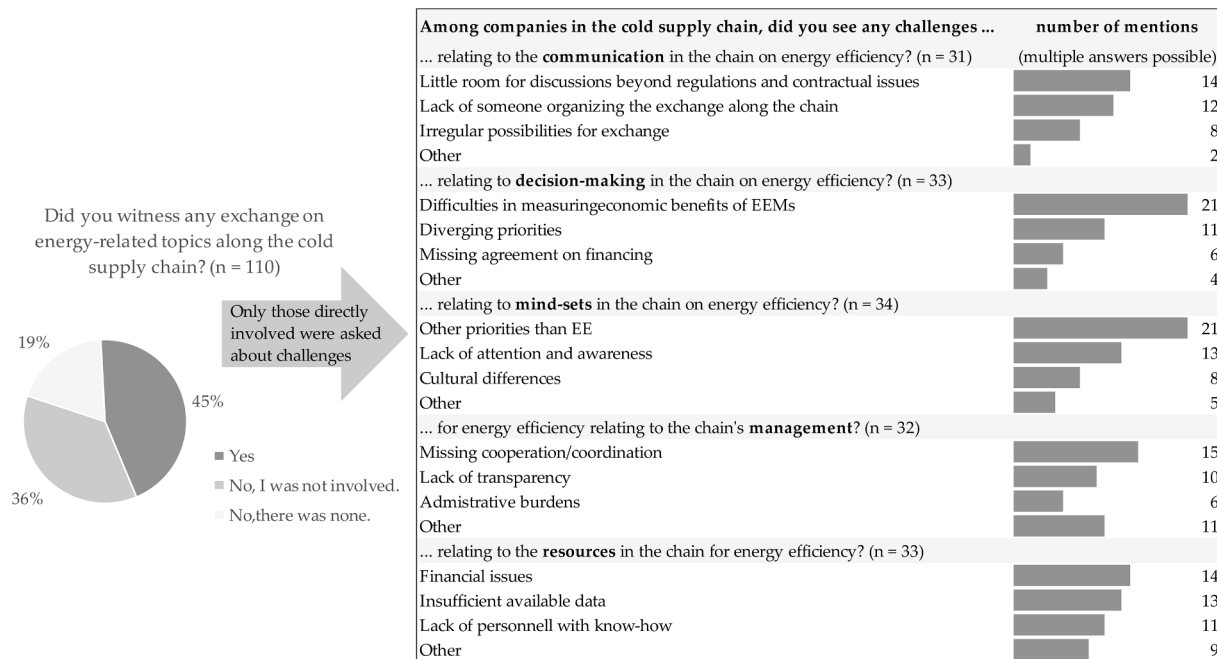


Fig. 8. Exchange on energy-related topics along the cold supply chain (left graph). Role of behavioural and organizational aspects for energy efficiency along the cold supply chain by category (communication, decision-making, mind-sets, management and resources (right table)).

also mentioned as a problem. The “others” category was mainly used for the categories “management” and “resources”.

To sum up, it can be observed that a substantial number of CSC members do not experience exchanges on energy-related issues. For those directly involved in the exchange, several challenges to CSC energy efficiency were identified at the organizational and behavioural levels.

5. Discussion

5.1. What role does energy efficiency play in CSCs of the food sector compared to its role in individual companies?

According to the interviews, companies in the CSC seem to agree that energy efficiency is a topic of high relevance, especially for refrigeration due to its substantial energy consumption and energy costs. From the perspective of both individual companies and entire CSCs, energy efficiency was claimed to be a relevant criterion in decision-making, even more so in the interviews than in the survey. The interviews also revealed that economic considerations dominate the exchange between companies; actual implementation of EEMs depends strongly on potential monetary advantages. The results also suggest that awareness about EEMs is lower along entire CSCs than within individual companies. In addition, no interviewee provided examples of specific EEMs implemented and coordinated together with other members of the CSC. Instead, organizations in the CSC seem to focus on their own situation and priorities, independence and flexibility.

The results in terms of the figures involved show that “storage and logistics” and “production and processing” play a crucial role in driving energy efficiency in the CSCs. This seems plausible for “storage and logistics”, which link all stages of the CSC. However, this result may be biased, as there was a high share of respondents from this particular group. Nevertheless, the results highlight that many stakeholders are not yet involved in driving this topic. Thus, energy efficiency could profit from more “people with real ambition” (Thollander and Palm, 2013) for energy efficiency in all areas of the CSCs.

To sum up, the importance of cooperating on energy efficiency in food CSCs is acknowledged, but implementing EEMs along CSCs is far

less common than doing so in individual companies. Fostering strategic cooperation on energy efficiency among CSC members seems to be an important preparatory step before/while dealing with the relational and regulatory barriers mentioned in the literature (such as in Centobelli et al., 2018).

5.2. What challenges do the structural features of CSCs pose for energy efficiency?

While the notion “from farm-to-fork” is appealingly simple, the survey underlines the heterogeneity of CSCs as is also evident in the literature (e.g. Centobelli et al., 2018; Majewski et al., 2020). CSCs in the food sector tend to be more often regional than global, and cooperation is characterised by long-term relationships with only a few newcomers/dropouts each year. This is in line with King et al. (2010), who point out that durable relationships between supply chain partners over time are important for all types of chains. However, relatively few companies in the CSCs seem to know the entire CSC well; individual links tend to know their direct suppliers and customers and those supplying and buying from these, but not the entire CSC. This is in line with the findings of Thomé et al. (2021), who conclude that supply chain relationships are dynamic in terms of size, shape and configuration, as well as invisible and potentially confidential. Such limited transparency can be a challenge when adopting EEMs, as it can complicate the exchanges and joint action along the CSC. In this line of thought, “smaller” CSCs might have the advantage that the companies already know each other. Similarly, it has been suggested in the literature that short food supply chains enjoy benefits such as close cooperation due to geographical and social proximity (Majewski et al., 2020). On the other hand, larger chains, possibly including many large companies with energy management systems, could benefit from sharing their knowledge and from a better exchange on energy-related issues. Marsden et al. (2000) also point out that long industrial chains are often complex but rationally organized. According to our survey, SMEs are also strongly represented in CSCs. As a result, the typical barriers to energy efficiency prevalent in SMEs (Sorrell et al., 2010; Thollander and Palm, 2013) can be expected in CSCs of the food sector as well. The results seem quite heterogeneous regarding the exchange of information in the CSCs beyond routine

exchanges: Half of the CSCs indicate exchanging information rather regularly, the others only at longer intervals (3+ months) or not at all. Rare exchanges might be detrimental to joint action on energy efficiency, since initiating and coordinating such activities might not gain sufficient momentum.

To sum up, the lack of transparency, complex structures and infrequent interaction on energy-related topics seem to be particularly relevant obstacles to energy efficiency in CSCs that come in addition to the barriers relevant in individual companies, especially SMEs. More detailed investigation of factors that positively/negatively influence energy efficiency in CSCs seems necessary, as also pointed out by [Cen-tobelli et al. \(2018\)](#).

5.3. What is the relevance of NEBs and how are they perceived in CSCs compared to individual companies?

NEBs besides savings in energy and carbon dioxide emissions are relevant for both individual companies and entire CSCs. The interview and survey results suggest that positive effects far outweigh any negative effects,² but the perception of NEBs along CSCs seems lower than in individual companies. A high number of respondents did not answer the question about the positive (or negative) effects of EEMs across CSCs. This suggests they had difficulties in conceiving of NEBs along CSCs and/or there was a lack of joint EEMs between several links in the CSCs. Where answers were provided, however, the categories of NEBs in the survey seemed to mirror relevant aspects well. Furthermore, the survey results hinted at differences in the relevance of particular NEBs between CSCs and individual companies. Production-related benefits such as improved operation and maintenance and improved productivity seem more important for individual companies, while improved image and marketing as well as waste reduction rank higher for CSCs. Emission reductions (excluding CO₂) seem to be important for both groups. During the interviews, however, the distinction between CSCs and individual companies was not very pronounced. This difference could stem from the open-ended question style and the fact that the interviews focused less on comparing NEB categories between individual companies and the supply chain.

It seems to be difficult for companies to measure the impacts of NEBs on individual companies and especially on CSCs. This lends support to the IEA's statement that "methodologies for quantifying wider benefits from EEMs in industry are still at the inception stage" ([International Energy Agency, 2014](#)). While the literature on NEBs shows that their strategic assessment can positively influence decisions on energy efficiency issues (e.g. [Cooremans, 2015](#)), further research is needed on approaches to assess and quantify such benefits across entire CSCs.

To sum up, the perception of NEBs along CSCs seems to be lower than in the individual companies. Focusing on NEBs can contribute to overcoming the difficulties related to implementing EEMs in CSCs, but this is only one part of a broader strategy that should also address the previously mentioned challenges to enhancing the energy performance in CSCs.

5.4. What are the particular challenges stemming from BOAs with regard to energy efficiency improvements along CSCs?

At the organizational and behavioural level of the CSC, it is evident from the interviews that energy is wasted due to a lack of coordination

and communication along the chain. This indicates that the cooperation of food companies in the field of sustainable or green supply chain management could be expanded even further, as outlined in the conceptual framework of [Govindan \(2018\)](#). Based on the statements of those directly witnessing exchanges on energy-related topics in CSCs, the survey highlights that the complexity of CSC operations is a challenge when implementing EEMs. Among other reasons, this is also due to the lack of a person responsible for organizing energy efficiency issues along the chain. In addition, behavioural and organizational challenges arise from different priorities and difficulties in measuring the economic benefits of EEMs as well as the lack of available data and financing issues. Other challenges include the lack of know-how and skilled personnel or the lack of attention to energy-related topics. Such barriers are also known from individual companies, but they might be more pronounced as the interaction in CSCs adds an additional layer of inter-organizational complexity to the adoption of EEMs.

To sum up, given the complexity of CSC's multiplex relations in the food sector ([Thomé et al., 2021](#)), it follows that behavioural and organizational aspects in their interaction are of particular relevance for the uptake of energy-efficiency measures and matter even more than in individual companies.

5.5. What are the implications of these findings for the theory on barriers to energy efficiency?

The taxonomy of barriers to energy efficiency includes six categories: risks, imperfect information, hidden costs, access to capital, split incentives and bounded rationality ([Sorrell et al., 2004](#)). As the numerous empirical investigations on barriers indicate (e.g. [Sorrell et al., 2010](#)), this taxonomy is applicable to many contexts, also because of its general character. Although the theory is developed with a focus on barriers within individual companies, our findings suggest that it is generally applicable also for inter-organizational structures such as the CSC. Yet the need for interactions on EEMs in CSCs adds an additional layer of inter-organizational complexity to the adoption of EEMs. The empirical results from this investigation suggest that the consideration of specific structural features and BOAs in CSCs is important to avoid neglecting important barriers. The findings can serve to operationalize the theoretical perspective on barriers by pointing out specific aspects beyond those relevant for individual companies that may become relevant in the multi-company settings as found in CSCs. [Table 1](#) provides an overview of such potential CSC specific barriers derived from the results of the investigation. While this is a starting point, further in-depth analyses beyond the exploratory analysis of conditions and perceptions are required to extend this work. In particular, this would concern a) further analyses of the relevance of individual barriers specific to CSCs, and b) whether some barriers are more prevalent in particular types of CSCs than others.

It can be observed that NEBs as a core element covered in this investigation have not been considered in [Table 1](#). This has several reasons: First, according to the results, only a negligible share has observed any negative impacts from implemented EEMs at all. Therefore, negative impacts do not seem to be of substantial relevance as barriers in CSCs. Second, the remaining (positive) benefits are expected to facilitate, not hinder, the adoption of EEMs in CSCs. And third, depending on the adopted perspective and the underlying definition, it can be argued that NEBs increase the economic attractiveness of EEMs if their impact is quantified. In the sense of the initially provided definition of barriers, they are thus affecting what is cost-effective, thereby increasing the potentially attractive measures that are subject to barriers.

² When explicitly asked about negative effects, participants in the interviews mentioned aspects such as additional personnel and financial resources (e.g. for qualified personnel or audits), difficulties with adapting to new systems causing potential frustration for workers or customers, and decreased productivity due to increased system unreliability and higher downtimes. Nevertheless, such negative effects generally play a minor role according to both the interviews and the survey.

Table 1

List of potential barriers to energy efficiency in CSCs requiring particular attention based on the empirical results from this investigation.

Structural features and BOAs in CSCs	Observation(s)/aspect(s)	Potential barrier specific to CSCs
1) Risk: Perceived technical or financial risks and/or uncertainty about businesses and markets		
Dimension	numerous companies and links are frequently involved in CSCs	perceived risk related to the ability to “control” the implementation of the EEMs in the CSCs (involvement of several parties takes part of the process out of the hand of individual actors)
Origin	some CSCs are operating on a global scale	perceived risk to accommodate for different legal setups/regimes present in the CSC
Familiarity	knowledge of actors in the CSCs is frequently limited	perceived risk of overlooking issues in unknown parts of the CSC that might affect the effectiveness of an EEM
Stability	companies in the CSC vary over time to some degree	perceived risk concerning the long-term viability of EEMs due to potential changes in the structure of the CSC
Mind-sets	cultural differences are present in some CSCs	perceived uncertainty related to cultural differences of actors in the CSCs which affect the implementation of EEMs
2) Imperfect information: Lack of information on energy efficient opportunities		
Dimension/ Familiarity/ Management	numerous companies and links are frequently involved in CSCs/ knowledge of actors in the CSCs is frequently limited/ there is a lack of transparency in the CSC	limited information quality and quantity due to the need of obtaining information from numerous involved links and companies in the CSC
Origin	some CSCs are operating on a global scale	lack of accessibility to information due to language barriers between actors in the CSC
Interaction	some actors in the CSC only rarely come in substantial contact with each other	limited opportunities for collecting information from other members of the CSC
Decision-making	difficulties in measuring economic benefits from EEMs are frequent in CSCs	challenging determination of potential savings and the associated economic benefits of EEMs for the CSC
Management	missing coordination and cooperation is a reoccurring issue in CSCs	missing coordination and cooperation across several actors of the CSC
Resources	available data is insufficient/personnel with know-how is missing	limited availability of know-how for the implementation of EEMs in the CSCs
3) Hidden costs: Perception of unaccounted costs		
Origin	some CSCs are operating on a global scale	cost associated with the compensation for differences in geographical location of the actors in the CSC (e.g. due to differences in legislation, standards, education, labour costs, culture)
Stability	companies in the CSC vary over time to some degree	costs associated with the preparation for and compensation of changes in the setups of CSCs
Communication/ Management	someone organizing the exchange along the CSC is frequently missing/	costs associated with the overall project coordination across all involved actors in the CSC

Table 1 (continued)

Structural features and BOAs in CSCs	Observation(s)/aspect(s)	Potential barrier specific to CSCs
	administrative burdens occur in some CSCs	
4) Access to capital: Limits in the availability of funding		
Dimension	numerous companies and links are frequently involved in CSCs	role of risks related to the required cooperation in the CSC for the availability of capital potentially lower financial commitment of individual companies due to burden sharing
Dimension	both large and small companies are typically involved in CSCs	differences in funding conditions for SMEs and non-SMEs involved in the CSC
Origin	some CSCs are operating on a global scale	availability of accessible capital in the different geographical locations of the actors in the CSC
Decision-making/ Resources	agreements on financing are missing in some cases/ financial issues occur frequently	missing financing agreements for EEMs among the members of the CSCs
5) Split incentives: Costs and benefits are attributed to different actors		
Dimension	numerous companies and links are frequently involved in CSCs	no available measurements and agreements on the distribution of costs and benefits from the EEMs among the involved actors in the CSC
6) Bounded rationality: Individual behaviour deviates from the decisions suggested by economic models		
Interaction/ Communication	the direct contact is limited in some CSCs/the scope for discussions beyond regulations and contractual issues is frequently limited	few opportunities for planning and establishing action of several actors in the CSC
Decision-Making/ Mind-sets	priorities are diverging in some cases/other priorities are frequently encountered/there is a lack of attention and awareness in some cases	central actors in the CSC with low awareness on EEMs attributing them a low priority

To sum up, the basic theoretical concept on barriers to energy efficiency could be operationalized with an additional layer of inter-organizational challenges. This is especially the case when investigations go beyond mere analyses on the relevance of barriers, but seek to investigate about rationales for interventions by energy policy making.

5.6. Limitations of the research

With regard to the interpretation of the findings, the general limitations of interview and survey approaches apply (Yin, 2009). First, they include a potential bias in the selection of participants, since only those interested in the topic tend to take part. Second, the interview/survey design, e.g. the order of questions, might affect the responses. Finally, the participants work in different areas of the CSCs, they have different cultural and technological backgrounds and operate under sometimes varying conditions, e.g. with regard to climate, energy prices and/or national policy measures. Especially concerning the interviews, the quality could vary due to different interviewers and translations. Emphasis was placed on ensuring a common understanding of the questions by providing a guideline with formalized questions to interviewees and additional separate instructions and examples for the interviewers. In addition, respondents might feel the need to provide

socially desirable answers in an interview situation. To minimize this, the focus was on factual information about existing processes rather than desirable situations. The online survey solves some of these problems, although there is obviously less control over the participants here. From a methodological point of view concerning BOAs, the suggested question scheme seems to match the perceived problems well, as the answer “other” reasons was not intensively used.

Where the results of the interviews and the survey differ, design differences must be acknowledged: While the interviews focused on initial exploration, the survey sought to pinpoint topics using closed questions. However, the overall statements regarding the role of energy efficiency, its underlying challenges as well as the lower awareness of this issue in CSCs point in the same direction and cut across different CSCs in the European food sector.

From a methodological perspective beyond the scope of this paper, future research could focus on particular segments of the food sector to rule out any intersectoral or geographical effects on the results, also by further validating the results using methods from analytical statistics such as multivariate or factor analysis beyond the descriptive approach used here.

6. Conclusions

The aim of this paper was to investigate the conditions for and perceptions of energy efficiency improvements in CSCs of the food sector and the implications for the theory on barriers to energy efficiency. While sustainable supply chain management is a widely discussed research stream from both theoretical (Seuring et al., 2022; Touboul et al., 2020) and practical perspectives (Liu et al., 2020; de Souza et al., 2022), cross-organizational aspects of energy efficiency and its practical implementation in particular have rarely been covered by empirical works. Thus, this exploratory analysis of energy efficiency in CSCs of the food sector contributes to understanding the cross-organizational dimension of energy efficiency. For this purpose, the structures of food CSCs were investigated and the role of NEBs and BOAs were analysed.

The results suggest that the importance of cooperating on energy efficiency in food CSCs is generally acknowledged, but implementing EEMs along CSCs is far less common than in individual companies. The main reason that emerges as particularly relevant for CSCs seems to be the lack of transparency and infrequent interactions on energy-related topics in CSCs. It was found that the need for cooperation among different companies on energy-related matters adds an additional layer of complexity to the practical adoption of EEMs in CSCs compared to the situation in individual companies. While this additional complexity can be analysed along the established lines of the theory on barriers to energy efficiency, it seems advisable to consciously take the structural features and BOAs in CSCs into account to avoid neglecting important barriers when analysing them. This seems especially relevant when elaborating strategies for overcoming barriers to achieving climate-neutrality of CSCs. While emphasizing NEBs, which are found to be less prominent in CSCs at present, is part of such strategies, addressing BOAs is of particular relevance for the uptake of EEMs there. For policy makers, this translates into a need for interventions that both address the inter- and intra-organisational level. Cooperation models on energy-related matters along CSCs broadening existing energy management approaches beyond individual organizations could be part of such strategies including methods on how to measure and share the costs and benefits of EEMs across CSCs.

7. Author Contributions

This paper is based on a completely revised and extended contribution by the authors to the ecece Industrial Summer Study (Neusel, L.; Hirzel, S.; Zanoni, S.; Marchi, B. ‘Energy efficiency from farm to fork? On the relevance of non-energy benefits and behavioural aspects along the cold supply chain’).

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Andrews, R.N., Johnson, E., 2016. Energy use, behavioral change, and business organizations: Reviewing recent findings and proposing a future research agenda. *Energy Res. Social Sci.* 11, 195–208. <https://doi.org/10.1016/j.erss.2015.09.001>.
- Banks, N., Redgrove, Z., 2012. In: *What Are the Factors influencing Energy Behaviour And Decision-making in the Non-domestic sector?: A Rapid Evidence Assessment*. University of Oxford, London, p. 88 pp..
- Brzozowska, A., Imiolczyk, J., Brzeszczak, A., Szymczyk, K., 2016. Managing cold supply chain. *IEEE International Conference on Advanced Logistics and Transport*.
- Cadilhon, J.-J., Fearn, A., Hughes, D., Moustier, P., 2003. Wholesale Markets and Food Distribution in Europe: New Strategies for Old Functions.
- Cagno, E., Moschetta, D., Trianni, A., 2019. Only non-energy benefits from the adoption of energy efficiency measures? A novel framework. *J. Cleaner Prod.* 212, 1319–1333. <https://doi.org/10.1016/j.jclepro.2018.12.049>.
- Carter, C.R., Rogers, D.S., 2008. A framework of sustainable supply chain management: moving toward new theory. *Int. Jnl. Phys. Dist. Log Manage* 38, 360–387. <https://doi.org/10.1108/09600030810882816>.
- Centobelli, P., Cerchione, R., Esposito, E., 2018. Environmental sustainability and energy-efficient supply chain management: A review of research trends and proposed guidelines. *Energies* 11, 275. <https://doi.org/10.3390/en11020275>.
- Chaudhuri, A., Dukovska-Popovska, I., Subramanian, N., Chan, H.K., Bai, R., 2018. Decision-making in cold chain logistics using data analytics: a literature review. *Int. Jnl. Logist. Manage.* 29, 839–861. <https://doi.org/10.1108/IJLM-03-2017-0059>.
- Chiffolleau, Y., Millet-Amrani, S., Canard, A., 2016. From short food supply chains to sustainable agriculture in urban food systems: food democracy as a vector of transition. *Agriculture* 6, 57. <https://doi.org/10.3390/agriculture6040057>.
- European Commission, 2018. A Clean Planet for all A European strategic long-term vision for a prosperous, modern, competitive and climate neutral economy. COM (2018) 773 final. <https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:52018DC0773> (accessed 19 May 2021).
- Cooremans, C., 2011. Make it strategic! Financial investment logic is not enough. *Energ. Eff.* 4, 473–492. <https://doi.org/10.1007/s12053-011-9125-7>.
- Cooremans, C., 2015. Competitiveness benefits of energy efficiency: a conceptual framework, 123–131.
- Dania, W.A.P., Xing, K., Amer, Y., 2018. Collaboration behavioural factors for sustainable agri-food supply chains: A systematic review. *J. Cleaner Prod.* 186, 851–864. <https://doi.org/10.1016/j.jclepro.2018.03.148>.
- de Souza, E.D., Kerber, J.C., Bouzon, M., Rodriguez, C., 2022. Performance evaluation of green logistics: Paving the way towards circular economy. *Cleaner Logist. Supply Chain* 3, 100019. <https://doi.org/10.1016/j.clscn.2021.100019>.
- DeCanio, S.J., 1998. The efficiency paradox: bureaucratic and organizational barriers to profitable energy-saving investments. *Energy Policy* 26, 441–454. [https://doi.org/10.1016/S0301-4215\(97\)00152-3](https://doi.org/10.1016/S0301-4215(97)00152-3).
- Diekmann, V., Kölle, A., Laumann, M., Geßner, C., 2015. Nachhaltigkeit in der FMCG-Branche: Eine Gemeinschaftsstudie des Zentrums für Nachhaltige Unternehmensführung und der Lebensmittel Zeitung, Witten, 54 pp.
- Eberle, U., Wenzig, J., Kölle, A., Geßner, C., Eichner, S., Gründler, L., 2016. Branchenleitfaden: Nachhaltigkeit in der Tiefkühlwirtschaft. Zentrum für Nachhaltige Unternehmensführung der Universität Witten/Herdecke, Berlin, p. 72.
- Elkington, J., 1994. Towards the sustainable corporation: win-win-win business strategies for sustainable development. *California Manage. Rev.* 36, 90–100. <https://doi.org/10.2307/41165746>.
- Elliott, J.A., 2013. *An introduction to sustainable development*, 4th ed. Routledge, London, p. 361.
- EUROSTAT, 2021. Energy Balances in the MS Excel File format (2021 edition): Energy balance sheets (nrg_bal_c). <https://ec.europa.eu/eurostat/documents/38154/4956218/Energy-balance-sheets-February-2021-edition.zip/4b1d6665-f303-be7d-a7e5-1e0da16ec0d9> (accessed 23 July 2021).

- Pawett, T., Killip, G., 2019. Re-thinking energy efficiency in European policy: Practitioners' use of 'multiple benefits' arguments. *J. Cleaner Prod.* 210, 1171–1179. <https://doi.org/10.1016/j.jclepro.2018.11.026>.
- Govindan, K., 2018. Sustainable consumption and production in the food supply chain: A conceptual framework. *Int. J. Prod. Econ.* 195, 419–431. <https://doi.org/10.1016/j.ijpe.2017.03.003>.
- Govindan, K., Shaw, M., Majumdar, A., 2021. Social sustainability tensions in multi-tier supply chain: A systematic literature review towards conceptual framework development. *J. Cleaner Prod.* 279, 123075 <https://doi.org/10.1016/j.jclepro.2020.123075>.
- International Energy Agency, 2014. *Capturing the Multiple Benefits of Energy Efficiency: A Guide to Quantifying the Value Added*. OECD Publishing, Paris, p. 232.
- Jaber, M.Y., Goyal, S.K., 2008. Coordinating a three-level supply chain with multiple suppliers, a vendor and multiple buyers. *Int. J. Prod. Econ.* 116, 95–103. <https://doi.org/10.1016/j.ijpe.2008.08.001>.
- Jaffe, A.B., Stavins, R.N., 1994. The energy-efficiency gap What does it mean? *Energy Policy* 22, 804–810. [https://doi.org/10.1016/0301-4215\(94\)90138-4](https://doi.org/10.1016/0301-4215(94)90138-4).
- James, S.J., James, C., 2010. The food cold-chain and climate change. *Food Res. Int.* 43, 1944–1956. <https://doi.org/10.1016/j.foodres.2010.02.001>.
- Khalid, R.U., Seuring, S., Wagner, R., 2020. Evaluating supply chain constructs in the base of the pyramid environment. *J. Cleaner Prod.* 270, 122415 <https://doi.org/10.1016/j.jclepro.2020.122415>.
- King, R., Hand, M., DiGiacomo, G., Clancy, K., Gomez, M., Hardesty, S., Lev, L., McLaughlin, E., 2010. Comparing the structure, size, and performance of local and mainstream food supply chains. *Economic Research Report 99*. Economic Research Service 81, pp.
- Li, L., Wang, B., Cook, D.P., 2014. Enhancing green supply chain initiatives via empty container reuse. *Transp. Res. Part E* 70, 190–204. <https://doi.org/10.1016/j.tre.2014.06.018>.
- Liu, Y., Zhu, Q., Seuring, S., 2020. New technologies in operations and supply chains: Implications for sustainability. *Int. J. Prod. Econ.* 229, 107889 <https://doi.org/10.1016/j.ijpe.2020.107889>.
- Majewski, E., Komarska, A., Kwiatkowski, J., Malak-Rawlikowska, A., Waś, A., Sulewski, P., Golaś, M., Pogodzińska, K., Lecoeur, J.-L., Tocco, B., Török, Á., Donati, M., Vittersø, G., 2020. Are short food supply chains more environmentally sustainable than long chains? A life cycle assessment (LCA) of the eco-efficiency of food chains in selected EU countries. *Energies* 13, 4853. <https://doi.org/10.3390/en13184853>.
- Malak-Rawlikowska, A., Majewski, E., Waś, A., Borgen, S.O., Csillag, P., Donati, M., Freeman, R., Hoàng, V., Lecoeur, J.-L., Mancini, M.C., Nguyen, A.n., Saïdi, M., Tocco, B., Török, Á., Veneziani, M., Vittersø, G., Wavresky, P., 2019. Measuring the economic, environmental, and social sustainability of short food supply chains. *Sustainability* 11 (15), 4004.
- Marchi, B., Zanoni, S., 2017. Supply chain management for improved energy efficiency: review and opportunities. *Energies* 10, 1618. <https://doi.org/10.3390/en10101618>.
- Marsden, T., Banks, J., Bristow, G., 2000. Food supply chain approaches: exploring their role in rural development. *Sociologia Ruralis* 40 (4), 424–438.
- Mikkola, M., 2008. Coordinative structures and development of food supply chains. *Br. Food J.* 110, 189–205. <https://doi.org/10.1108/00070700810849907>.
- Mills, E., Rosenfeld, A., 1996. Consumer non-energy benefits as a motivation for making energy-efficiency improvements. *Energy* 21, 707–720. [https://doi.org/10.1016/0360-5442\(96\)00005-9](https://doi.org/10.1016/0360-5442(96)00005-9).
- Monczka, R.M., 2009. *Purchasing and supply chain management*, 4th ed. South-Western, Mason, OH, p. 810.
- Monforti-Ferrario, F., Dallemand, J.-F., Pinedo Pascua, I., Motola, V., Banja, M., Scarlat, N., Mederac, H., Castellazzi, L., Labanca, N., Bertoldi, P., et al., 2015. *Energy use in the EU food sector: State of play and opportunities for improvement*. JRC Sci. Policy Report. Joint Research Centre, Luxembourg, 1 online resource.
- Murphy, J., Adair, P., 2013. Sustainable supply chain management in the Food Sector 27, 37–40.
- Neusel, L., Hirzel, S., Zanoni, S., Marchi, B., 2020. Energy efficiency from farm to fork? On the relevance of non-energy benefits and behavioural aspects along the cold supply chain. In: *Eceee Industrial Summer Study Proceedings*, pp. 101–110.
- Palazzo, J., 2009. Placing barriers to industrial energy efficiency in a social context: a discussion of lifestyle categorisation. *Energy. Eff.* 2, 263–270. <https://doi.org/10.1007/s12053-009-9042-1>.
- European Parliament, 2013. Regulation (EU) No 1305/2013 of the European Parliament and of the Council of 17 December 2013 on Support for Rural Development by the European Agricultural Fund for Rural Development (EAFRD) and Repealing Council Regulation (EC) No 1698/2005, Brussels, Belgium, 62 pp.
- Ryan, L., Nina, C., 2012. *Spreading the Net: The Multiple Benefits of Energy Efficiency Improvements* 08. IEA Energy Papers, Paris.
- Saldana, J., 2011. *The Coding Manual for Qualitative Researchers*. Sage, London.
- Seuring, S., Aman, S., Hettiarachchi, B.D., de Lima, F.A., Schilling, L., Sudasinghe, J.L., 2022. Reflecting on theory development in sustainable supply chain management. *Cleaner Logist. Supply Chain* 3, 100016. <https://doi.org/10.1016/j.clscn.2021.100016>.
- Seuring, S., Müller, M., 2008. From a literature review to a conceptual framework for sustainable supply chain management. *J. Cleaner Prod.* 16, 1699–1710. <https://doi.org/10.1016/j.jclepro.2008.04.020>.
- Shashi, S., Cerchione, R., Singh, R., Centobelli, P., Shabani, A., 2018. Food cold chain management. *Int. J. Logist. Manage.* 29, 792–821. <https://doi.org/10.1108/IJLM-01-2017-0007>.
- Smith, K.M., Wilson, S., Hassall, M.E., 2021. Could focusing on barriers to industrial energy efficiency create a new barrier to energy efficiency? *J. Cleaner Prod.* 310, 127387.
- Sorrell, S., Mallett, A., Nye, S., 2010. *Barriers to Industrial Energy Efficiency: A Literature Review: Background Study for the UNIDO Industrial Development Report (IDR) 'Industrial Energy Efficiency Pays, Why Is It Not Happening?'*. European Commission - Non Nuclear Energy Programme: Joule III, JOS3CT970022, Brighton.
- Sorrell, S., O'Malley, E.J., Schleich, J., Scott, S., 2004. *The economics of energy efficiency: Barriers to cost-effective investment*. Elgar, Cheltenham, p. 349.
- Srivastava, S.K., 2007. Green supply-chain management: A state-of-the-art literature review. *Int. J. Manage. Rev.* 9, 53–80. <https://doi.org/10.1111/j.1468-2370.2007.00202.x>.
- Thema, J., Suerkemper, F., Couder, J., Mzavanadze, N., Chatterjee, S., Teubler, J., Thomas, S., Ürges-Vorsatz, D., Hansen, M.B., Bouzarovski, S., Rasch, J., Wilke, S., 2019. The multiple benefits of the 2030 EU energy efficiency potential. *Energies* 12, 2798. <https://doi.org/10.3390/en12142798>.
- Thollander, P., Karlsson, M., Rohdin, P., Wollin, J., Rosenqvist, J., 2020. *Barriers to energy efficiency*. *Introd. Indus. Energy Eff.* Elsevier 291–305.
- Thollander, P., Palm, J., 2013. *Improving Energy Efficiency in Industrial Energy Systems*. Springer, London, London.
- Thomé, K.M., Cappellesso, G., Ramos, E.L.A., Duarte, S.C.d.L., 2021. Food Supply Chains and Short Food Supply Chains: Coexistence conceptual framework. *Journal of Cleaner Production* 278, 123207. <https://doi.org/10.1016/j.jclepro.2020.123207>.
- Touboulic, A., McCarthy, L., Matthews, L., 2020. Re-imagining supply chain challenges through critical engaged research. *J. Supply Chain Manag.* 56, 36–51. <https://doi.org/10.1111/jscm.12226>.
- Trianni, A., Cagno, E., Thollander, P., Backlund, S., 2013. Barriers to industrial energy efficiency in foundries: a European comparison. *J. Cleaner Prod.* 40, 161–176. <https://doi.org/10.1016/j.jclepro.2012.08.040>.
- United Nations, 2015. Paris Agreement.
- Worrell, E., Laitner, J.A., Ruth, M., Finnan, H., 2003. Productivity benefits of industrial energy efficiency measures. *Energy* 28, 1081–1098. [https://doi.org/10.10](https://doi.org/10.1016/S0360-5442(03)00091-4)